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Financial distress Prediction based on deep learning model

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Abstract

In order to better solve the problem that the traditional prediction model can not effectively describe the causes of enterprise financial crisis, this article uses deep learning model to further improve the effectiveness of enterprise financial distress prediction. To this end, this article selects A-share (non-financial industry) listed companies in Shanghai and Shenzhen from 2001 to 2018 as ST sample companies, the financial distress prediction model of listed companies including 23 financial indicators based on deep learning was constructed, and the BP algorithm was used to test and verify the model. The prediction accuracy of the model is 78.33%, and the accuracy of predicting ST company is 82%, which verifies the feasibility of the model in predicting financial distress.

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Keywords: deep learning; Listed company; Prediction of financial distress

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1. Introduction

Enterprises will face many risks in the process of development, and financial risk is an inevitable type of risk. Once any enterprise has a financial crisis, it will inevitably have a certain impact on the interests of the company, investors and the stability of the capital market. In the face of increasingly fierce market competition, on the one hand, enterprises in this environment will have a broader space for development, on the other hand, there will also be a lot of crises, once the financial crisis occurs, it is likely to lead to bankruptcy of enterprises. Most of the companies will face a gradual deterioration of the financial crisis. Therefore, if we can timely find the problems in the process of enterprise operation, timely predict the financial crisis signals and possible financial losses of enterprises, and ensure that enterprises can timely make effective adjustments to the actual situation in the

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embryonic stage of financial crisis, we can better realize the healthy development of enterprises. Based on this, this article constructs a deep learning model based on financial distress prediction, and takes listed companies as an example to better capture the financial crisis that may exist in the development process of enterprises through the improvement of the model, and timely help enterprises stop loss [1].

2. Literature review

For some analysts, recession is a model for determining whether a company is in financial trouble, and self-determination is a model for predicting stress finance first [2]. Based on the single financial ratio data of 19 sample companies, the financial status of the company is predicted. The warning value of financial indicators is set to determine whether the enterprise has fallen into financial distress. At the same time, it was determined that revenues (membership equity and membership equity) had the greatest impact on the results observed [3]. Based on data from the financial assessments of 79 companies in and out of financial crisis, an estimate of 30 financial indicators was assessed using a different determination. The results showed that the estimation of the value of the income / total expenses index was the best and the model had accurate accuracy. However, univariate analysis method can only use one financial ratio index to predict the financial situation, so it is difficult to determine which index can effectively distinguish the financial normal company from the financial distress company, and the model is set too simple, so its forecasting ability is very limited[4]. Based on the non-matching principle, the distribution of sample companies on the probability interval of bankruptcy, the relationship between the two discriminant errors and the segmentation points are studied by using the Logistic nonlinear regression model. The results show that there are no less than four kinds of indicators which can greatly affect the probability of bankruptcy of enterprises, and the prediction accuracy of bankruptcy can be more than 90% by using these four variables. Through F test on the financial index data of bankrupt companies and non-bankrupt companies, the results show that financial leverage, rate of return, stock return and fixed income security can significantly affect the accuracy of the model prediction. However, this method still has the constraint that collinearity cannot exist between independent variables[5]. Based on 70 financially distressed enterprises and 70 financially normal enterprises as research samples, Domestic researchers have successfully used Fisher's linear decision analysis, various horizontal modeling, and logistic regression analysis to estimate a company's financial position. By comparing the prediction results of the three methods, the Logistic regression model has the best prediction effect. It is pointed out that the financial forecast using cash flow index is short-term, and the linear probability model and Logistic model are used to predict the financial situation respectively. The conclusion is that the earning cash ratio is more important in both models, and the Logistic model has better predictive ability[6].

3. Construction of financial distress prediction of listed companies based on deep learning model

3.1. Deep learning model construction

The purpose of this line is to create a multi-layer deep neural network model that predicts the financial problems of listed companies. As shown in Figure 1: The bottom layer is the input layer, and the input file is the financial indicator selected in this section. An article that estimates the financial situation of listed companies. The number of d cells in the entry process is determined by the final number of options. The process layer is the latent layer, the k layer is the latent layer, and the number of neurons in each layer is determined only after the training network. The top procedure is the output procedure, which is the result of the final distribution [7]. The result is the result that the listed companies will be divided into ST and NON-ST, respectively.

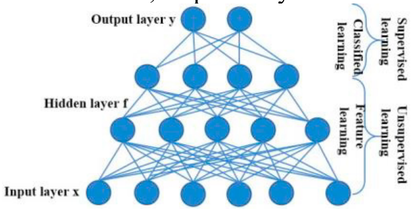

Figure 1 Deep learning model

This article will select as many financial indicators as possible that can fully reflect the financial situation of the enterprise, reduce the dimensionality of the multidimensional financial indicator data of the company through unsupervised learning (feature learning), extract useful features, and then classify the sample company through supervised learning (whether it is ST). Given training set is shown in Formula (1):

$$D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_m, y_m)\}, x_i \in R^d, y_i \in R^2 \quad (1)$$

m represents the number of samples, x represents the d -dimensional vector, representing d selected financial indicators, and y represents the 2-dimensional vector, representing two different classification results (ST and NON-ST) [13]. The selected financial indicators are mapped to the hidden layer, and the output results of each neuron in the first hidden layer, as shown in Formula (2):

$$F_{11} = f\left(\sum_{i=1}^{d} \omega_{i1} x_i + b_1\right)$$
$$F_{12} = f\left(\sum_{i=1}^{d} \omega_{i2} x_i + b_2\right)$$
$$\cdots$$
$$F_{1p} = f\left(\sum_{i=1}^{d} \omega_{ip} x_i + b_p\right) \quad (2)$$

The output of the first latent layer is then taken as input from the second latent layer, and the output of each neuron of the second latent layer, as shown in Formula (3):

$$F_{21} = f\left(\sum_{i=1}^{p} v_{i1} F_{1i} + c_1\right)$$
$$F_{22} = f\left(\sum_{i=1}^{p} v_{i2} F_{1i} + c_2\right)$$
$$\cdots$$
$$F_{2q} = f\left(\sum_{i=1}^{p} v_{iq} F_{1i} + c_q\right) \quad (3)$$

3.2. Selection of research objects

This line will select both good and bad companies with bad financial institutions and NON-ST companies. When selecting a timeline survey, this article will select financial statements (t-3) of all company models for t year to estimate whether the company will be ST [8]. The reason why according to the process involved in the publication of the annual publications of listed companies in the Company of China, the deadline for the publication of the publications The annual period of listed companies is within 4 months after the end of the annual financial year. Therefore, the time to identify the company in t-year ST and (t-1) annual reports is almost the same. Therefore, it is not advisable to use the information in the annual report of the company (t-1) to estimate whether the company will be estimated by ST year t. This undoubtedly exaggerates the predictive power of the model and has no effect on prior prediction. Therefore, the financial data of all sample companies (t-3) will be selected in this article to predict whether the company will be ST in t year, so as to ensure that the trained model can be objective and robust in achieving high prediction accuracy [9]. When selecting ST and NON-ST sample companies, the following criteria are mainly followed:

In t year, whether ST company or NON-ST company is selected, the financial data of (t-3) year is needed to predict the situation of T year. Therefore, it is necessary to ensure that all companies in the sample of t years have been listed at least in (t-3) years, otherwise it is impossible or difficult to obtain financial indicator data.

(1) Due to the particularity of financial companies, financial companies are excluded from the sample companies in this study;

(2) When selecting the sample of ST companies, only the companies that are ST for the first time are selected, that is, the companies change from normal to ST state for the first time since listing. For example, a company was ST for the first time in 2001, and then became normal and experienced ST for many times, then only the financial data of the company in 1998 were selected to predict the state of the company in 2001.

(3) When selecting the sample of ST company, other companies that are not in abnormal financial condition are

excluded, so that the sample can better match the research content of this article;

(4) When selecting the sample of NON-ST companies, we select the NON-ST companies of the current year with a certain proportion of the number of companies selected for the first time every year. NON-ST companies here refer to companies that have never been ST during the time span studied in this article [10].

3.3. Establishment of index system

According to the previous experience and the comprehensive principle of index selection, this article initially selected the above 27 second-level indicators. In the process of data collection, there are a lot of missing data of the four financial indicators including interest multiple, expansion rate of fixed asset investment, net cash content of net profit and cash dividend guarantee multiple. According to the operable principle of index selection, these four indicators should either be eliminated in order to retain all sample companies, or the sample companies with missing data should be eliminated in order to retain these four indicators. Since there are not many ST sample companies, this article keeps these sample companies and removes the above four indicators. Therefore, the index system finally determined in this article consists of six first-level indicators and twenty-three second-level indicators, as shown in Table 1.

Table 1 Finalized index system

First-level index	Second-level index	Calculation formula
Solvency indicator	Flow ratio X1	Current assets/current liabilities
	Quick ratio X2	(Current assets - net inventories)/ current liabilities
	Cash ratio X3	(Monetary capital + tradable financial assets + notes receivable)/ current liabilities
	Equity ratio X4	Total liabilities/Total shareholders' equity
	Cash flow liability ratio X5	Net cash flows/current liabilities from operating activities
Profitability indicator	Return on equity X6	Net profit attributable to shareholders of parent company/average shareholders' equity ×100%
	Return on total assets X7	Earnings before interest and tax (EBIT)/ Average total assets ×100%
	Net interest rate on total assets X8	Net profit (including profit and loss of minority shareholders)/ total average assets ×100%
	Net profit margin on sales X9	Net profit/operating revenue ×100%
	Operating margin X10	Operating profit/operating revenue ×100%
	Earnings per share X11	Current earnings / weighted average of current shareholders
Operational capability indicator	Inventory turnover X12	Operating cost/average net inventory
	Accounts receivable turnover X13	Operating income/average net accounts receivable
	Fixed asset turnover rate X14	Operating income/average total fixed assets
	Current asset turnover X15	Operating income/Total average current assets
	Total asset turnover X16	Operating income/Total average assets
Indicators of growth ability	Net profit growth rate X17	(Net profit for the same period - net profit for the same period last year) / Net profit for the same period last year × 100%
	Growth rate of total assets X18	(Total assets at the end of time - all assets at the beginning of the reporting period) / Total assets at the beginning of time × 100%
	Revenue growth rate X19	(Current earnings - earnings for the same period last year) / Earnings for the same period last year × 100%
Cash flow indicator	Cash operating index X20	Net cash flow from operating activities/gross operating cash flow
	Cash recovery rate of total assets X21	Net cash flow from operating activities/Total ending assets ×100%
	Net operating income cash content X22	Net cash flow from operating activities/operating income
Capital structure indicator	Asset-liability ratio X23	Total liabilities/Total assets x 1100%

4. A study of financial stress estimates of listed companies based on in-depth research standards

4.1. Sample Selection

In this article, A total of 465 listed companies in Shanghai and Shenzhen A-shares (non-financial industries) from 2001 to 2018 were selected as ST sample companies because of their abnormal financial conditions and were specially treated for the first time. A total of 2325 companies that have never been ST from 2001 to 2018 were selected as NON-ST sample companies with 5 times the number of ST companies each year.

In addition, when NON-ST sample companies are selected every year, the overall number of selected companies is 5 times that of ST sample companies. This article also makes statistics on the number of companies in various industries (non-financial industries) that are first ST listed every year, and then selects samples of NON-ST companies in corresponding industries with the number of ST companies in each industry 5 times.

4.2. Financial distress prediction of listed companies based on deep learning network model

(1) Training of deep learning network model

After many times of network training and debugging, this article finally determines the network structure that is suitable for the prediction effect of this study. It contains four hidden layers, and the dimensions of each layer are 18,14,10 and 5 respectively. Therefore, this article finally builds a five-layer deep learning network (23,18,14,10,5,2). In the first step, auto-encoder algorithm is used for feature learning, and in the second step, BP algorithm is used for classification learning. Next, the training process and results will be shown in detail.

Firstly, the first four layers of deep learning network (23,18,14,10,5) are trained unsupervised layer by layer by layer by auto-encoder algorithm. First, the first Encoder1 (23,18) and decoder Decoder1 (18,23) were trained. The training loss function figure is shown in figure 2.

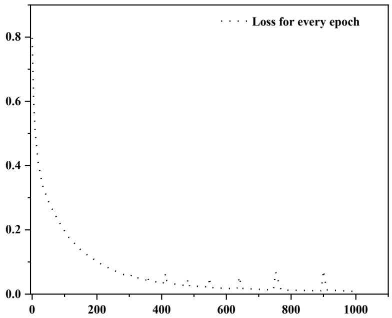

Figure 2 Root mean square error of first-layer parameter training in deep learning network

It can be seen from Figure 2 that when the training times are less than 400, the mean square error loss function decreases rapidly. After 400 times, the value of the loss function began to decline steadily, accompanied by a small range of local oscillation. But in general, the loss function shows a downward trend and eventually converges [19].

The second layer parameter training loss function is shown in Figure 3. In this round of training, the mean square error MSE loss function was used to calculate the reconstruction error, and the Tanh function was used as the nonlinear activation function. After repeated debugging, the learning rate of this round of training is set at 0.001, which can get better training effect. It can be seen that when the number of training is less than 200 times, the mean square error loss function decreases rapidly. After 200 times, the value of the loss function decreases slowly, and finally tends to be stable, and the loss function reaches convergence.

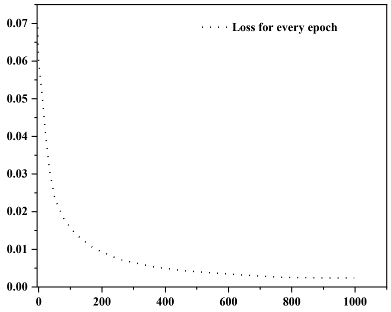

Figure 3 Root mean square error of layer 2 parameter training in deep learning network

After repeated debugging, the learning rate of this round of training is set at 0.001, which can get better training effect. As shown in Figure 4, the mean square error loss function drops rapidly when the training times are less than 100 times. From 100 to 400 times, the cost of labor decreases rapidly. After 400 training sessions, the unemployment gradually started to decline, eventually reaching a similar level.

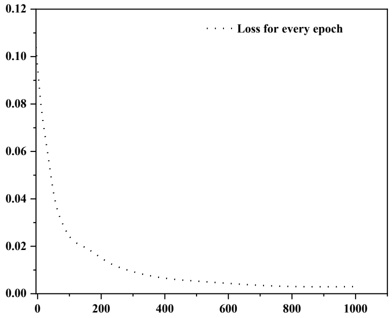

Figure 4 Root mean square error of layer 3 parameter training in deep learning network

After completing the third layer of training, Encoder3 and Decoder3 are saved. The Code2 output after the second encoder is input to the trained Encoder3 (14,10), and Code3 is output, which is then used as the input of the fourth Encoder4 (10,5) and decoder Decoder4 (5,10) for the fourth layer parameter training. The loss function of training is shown in Figure 5.

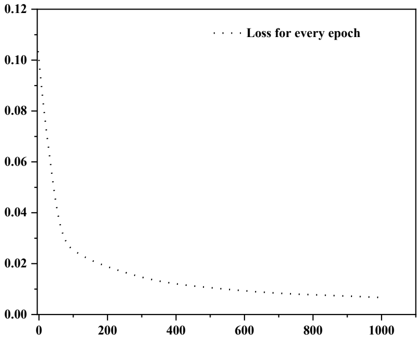

Figure 5 Root mean square error of layer 4 parameter training in deep learning network

Verification of deep learning network model

The test set contains a total of 300 samples, including 50 ST companies and 250 NON-ST companies. The samples of the test set do not coincide with the samples of the training set. The financial indicator data of the sample company in the test set (T-3) in year T are successively passed through Encoder1, Encoder2, Encoder3, Encoder4 and Classifier, and the financial state prediction results of year T are shown in Table 2:

Table 2 Prediction results of deep learning network model

The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	0 0 0 0 0 1 0 1 0 0 0 0 0 0 1 0 0 0 1 0 0 1 0 1 0 0 0 1 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0
The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	0 0 0 0 0 1 1 0 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 1 1 1 0 0 0 0 0 0 0 1 0 1 0 0 0 0 1 0 1 0 0
The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	1 1 1 0 0 1 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 1 1 0 1 1 0 1 0 0 0 0 0 0 1 0 1
The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 1 0 0 0 1 0 0 0 1 0 0 1 0 0 0 0
The actual	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	1 1 1 1 1 1 *0* 1 1 *0* *0* 1 1 1 *0* 1 *0* 1 1 1 1 1 1 1 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 1 0
The actual	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	1 1 1 1 1 1 1 1 *0* 1 1 *0* *0* 1 1 1 1 1 *0* 1 1 1 1 1 1 0 0 1 1 1 0 1 0 1 1 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0

0 represents the NON-ST company, 1 represents the ST company, the black number indicates that the model correctly predicts the financial state of the company and the red number indicates that the model incorrectly predicts the ST company to be the NON-ST company [20]. The confusion matrix between the actual state and the predicted state was calculated by statistical calculation of the predicted results, so as to calculate the overall accuracy and sub-classification accuracy of the deep learning model, as shown in Table 3:

Table 3 Prediction accuracy of deep learning network model

Prediction state	Actual state	NON-ST Company	ST company	Accuracy of prediction
NON-ST Company		194	56	77.60%
ST company		9	41	82.00%
				78.33%

Table 3 shows that in the view of NON-ST companies, the standard to be identified was 194 NON-ST companies and 56 unconvented NON-ST companies. sold as ST companies. The accuracy of NON-ST company estimates can be estimated at 77.60%. In ST company thinking, the model was identified as 41 ST company. The accuracy of ST company assumptions is 82.00%, as nine ST companies have decided not to be ST companies. On the whole, the model correctly identified 235 company states, with an overall prediction accuracy of 78.33%.

First, the Logistic model is trained. In order to make an effective comparison, the same training set is used to train the Logistic (23,2) binary classification model. The cross-entropy loss function of the training is shown in figure (7): it can be seen that the value of the cross-entropy loss function basically converges after 1000 times of training. Similarly, the test set will be used to verify the generalization ability of the Logistic model. The prediction results of the Logistic model are shown in Table 4.

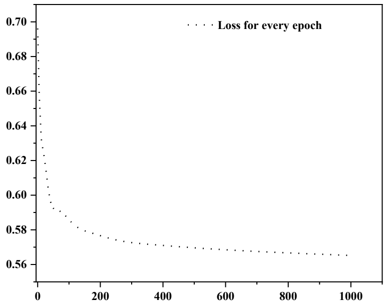

Figure 6 Cross entropy loss function of Logistic model training

Table 4 Prediction results of Logistic model

The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	0 0 0 0 0 1 0 1 0 1 0 0 0 1 1 0 0 0 1 0 0 0 0 1 0 0 0 1 0 0 1 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0
The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	0 0 1 0 0 1 1 0 0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 1 1 1 1 1 0 0 0 1 0 0 0 1 0 0 0 0 0 0 1 0 1 0 0
The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	1 1 1 1 1 0 1 1 0 0 0 1 0 1 0 1 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 1 1 0 1 1 0 1 0 0 0 0 0 0 1 0 1
The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	0 0 0 0 0 1 1 1 0 0 0 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 1 1 1 1 0 0 0 0 0 0 0 1 0 0 1 1 0 0 0
The actual	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	1 1 1 *0* 1 *0* *0* 1 1 *0* *0* 1 1 *0* *0* 1 *0* 1 1 1 1 1 1 1 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0
The actual	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	1 1 1 1 1 *0* 1 1 *0* 1 1 *0* *0* 1 1 1 1 1 *0* 1 1 1 1 1 1 1 0 1 1 1 0 1 0 1 1 1 1 1 1 0 1 1 0 0 1 0 0 0 1 0

The confusion matrix between the actual state and the predicted state was calculated statistically based on the prediction results, and the overall accuracy and sub-classification accuracy of the Logistic model were calculated as shown in Table 5:

Table 5 Prediction accuracy of Logistic model

Prediction state	Actual state	NON-ST Company	ST company	Accuracy of prediction
NON-ST Company		176	74	70.40%
ST company		13	37	74.00%
				71.00%

The Logistic model trained in this article has an overall prediction accuracy of more than 70% for financial distress of listed companies, but its performance is obviously inferior to that of the deep learning model. This further proves that establishing deep model can effectively improve the expression ability of the model.

4.3. Financial distress prediction of listed companies based on BP neural network model

In this article, two steps of auto-encoder unsupervised training and BP supervised training are used to train the deep learning model in order to improve the defect of training the deep neural network only with BP algorithm. In order to verify this point, BP algorithm will be adopted in this section to conduct supervisory training on the fifth layer of deep neural network (23,18,14,10,5,2). The loss function of training is shown in Figure 7.

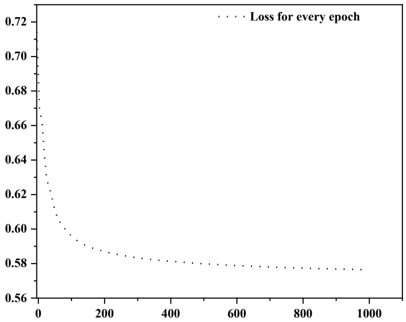

Figure 7 Cross entropy loss function of BP neural network model training

After the model training is completed, the test set is put into the trained model, and the prediction results are shown in Table 6.

Table 6 Prediction results of BP neural network model

The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	0 0 0 0 0 1 0 1 0 1 0 0 0 1 1 0 0 0 1 0 0 0 0 1 0 0 0 1 0 0 1 0 0 1 0 0 0 0 1 1 0 0 0 0 0 1 0 0 0 1
The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Prediction	0 0 1 0 0 1 1 0 1 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1 1 1 0 1 1 0 0 0 1 0 0 0 1 0 0 0 0 0 0 1 0 1 0 0
The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	0 0 1 1 0 1 1 0 0 0 1 0 1 0 1 0 0 0 0 0 1 0 0 1 0 0 0 1 0 0 0 0 0 0 1 1 0 1 1 0 1 0 0 0 0 0 0 1 0 1
The actual	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	0 0 0 0 0 1 1 1 0 0 0 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 1 1 1 1 0 0 0 0 0 0 0 1 0 0 1 1 0 0 0
The actual	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	1 1 1 *0* 1 *0 0* 1 1 *0 0* 1 1 *0 0* 1 1 1 1 1 1 *0* 1 1 1 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0 0 0 0
The actual	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Prediction	*0* 1 1 1 1 *0* 1 1 *0 0* 1 *0* 1 1 1 1 1 1 *0* 1 1 1 1 *0* 1 1 0 1 1 1 0 1 0 1 1 1 1 0 0 0 1 1 0 0 1 0 0 0 1 0

The confusion matrix between the actual state and the predicted state was calculated by statistical calculation of the prediction results, and the overall accuracy and sub-classification accuracy of the BP neural network model were calculated as shown in Table 7:

Table 7 Prediction accuracy of BP neural network model

Prediction state	Actual state	NON-ST Company	ST company	Accuracy of prediction
NON-ST Company		174	76	69.60%
ST company		15	35	70.00%
				69.67%

4.4. Comparison and analysis of prediction results

In this article, the deep learning model, Logistic model and BP neural network model are respectively used to predict the financial state of the listed company (T-3) in T year based on its financial index data. The prediction results are shown in Table 8:

Table 8 Comparison of prediction results

	Deep learning model			Logistic model			BP neural network model		
	Judge correctly	Judge wrongly	Accuracy Rate	Judge correctly	Judge wrongly	Accuracy Rate	Judge correctly	Judge wrongly	Accuracy Rate
NON-ST	194	56	77.60%	176	74	70.40%	174	76	69.60%
ST	41	9	82.00%	37	13	74.00%	35	15	70.00%
Total	235	65	78.33%	213	87	71.00%	209	91	69.67%

From the comparison results of the three models, it can be seen that the deep learning model performs better than the Logistic model and BP neural network model in predicting NON-ST companies and predicting ST companies, as well as in the overall forecasting ability of the model.

A comparison of the benefits of in-depth training models and logistics model needs suggests that the overall approximation of in-depth training models is better than in logistics models. The accuracy of the first assumption is 78.33%, and the last is 71.00%. Comparing the assumptions of the two models, making the depth of the model better the model's ability to describe the real world, and this improvement in presentation is mainly due to the performance of the model in-depth learning standards.

The NON-ST company is 77.60% accurate and the logistics model is 70.40% accurate. Compared with the logistics model, the in-depth research model has similarities in estimating NON-ST companies. In ST estimates, in-depth research models show higher estimates than delivery models, with estimated accuracy of 82.00% and 74.00%. In practice, the higher the accuracy of ST requirements, the lower the negative understanding, because the cost of reporting ST-based non-CT companies is greater than the NON-ST. According to the company ST, ST's high-precision (82%) and low error (18%) in the depth of the study model have the advantage of learning about financial difficulties. Comparing the resulting estimates of the deep learning model and the BP neural network model, we can see that the total estimates of the deep learning model, the NON-ST project, and the ST the forecast was better than before. The accuracy of the total BP neural estimates was 78.33% and 69.67%, respectively. When it comes to predicting NON-ST companies, the prediction accuracy is 77.60% and 69.60%, respectively. When predicting ST company, the accuracy of the two predictions is 82.00% and 70.00% respectively. It can be seen from the comparison of the prediction results of the two models that although the network structure of the two models is the same (23,18,14,10,5,2) five-layer network, the prediction ability of the models is greatly different due to different training algorithms. It can be seen from this that the deep learning model is not only able to show stronger prediction ability by simply deepening the layers of traditional shallow neural network, but also depends on the improvement of training algorithm to a large extent. In the previous article, it has been mentioned that BP algorithm is suitable for training shallow neural networks. As the number of layers deepens, the training of BP algorithm has become very difficult. If BP algorithm is used to train a deep neural network, its prediction ability may be inferior to that of shallow neural network.

Therefore, in this article, the auto-encoder algorithm is first used for unsupervised layer by layer greedy training, and then the BP algorithm is used for supervised training to train the whole network. This not only solves the problem of gradient diffusion of the BP algorithm to train all the deep gradients, but also improves the distribution of the samples. Comparing the benefits of standard logistics model and BP neural network model, the overall accuracy of the standard logistics model is slightly higher than that of BP neural network model, while the overall accuracy of the logistics model and BP neural network model standard is slightly higher. The patterns are 71.00% and 69.67%. NON-ST estimated accuracy of Logsitic model at 70.40% and accuracy of BP neural network model at 69.60%. ST's calculations are accurate at 74.00% and 70.00%, respectively. Comparing the hypotheses of the two models, it can be argued that simply increasing the neural network lag does not work better than competition in the network, network issues, train the deep neural network.

5. Conclusion

Compared with the traditional financial distress prediction model, the deep learning model can depict the real world from linear to nonlinear and from shallow learning to deep learning. This improvement enables the deep learning model to depict the nonlinear complex relationship between high-dimensional data and the hidden features, and finally achieve the purpose of feature judgment. Based on this, this article constructs a five-layer (23,18,14,10,5,2) deep learning network model, using the 23-dimensional financial index data of Shanghai and Shenzhen A-share (non-financial) listed companies (T-3) in t years to predict their financial status (whether ST). From the results, the deep learning model is obviously superior to the Logistic model and BP neural network model in both the ability of predicting ST companies and NON-ST companies and the overall forecasting ability of the model. Especially in the prediction of ST Company, the prediction accuracy of the deep learning model is as high as 82%. Which indicates that the deep learning model constructed and trained in this article can be fully applied to the financial distress prediction of listed companies. Moreover, compared with the traditional Logistic model and BP neural network model, it has a stronger ability of three years' early warning.

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